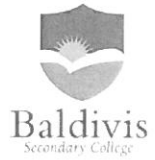


Year 9 Mathematics Test
End of Term Test- Term 1
Resource Free Section (No calculators, no notes)



Time: 10 mins Name: _____



/10 marks

Multiple Choice Section: Please circle the correct answer (only circle one)

1. Which of the following is equal to the expression $a \times 7 \times 8 \times b \times a$?

- A) $56ab^2$
- ☒ B) $56a^2b$
- C) $63ab^2$
- D) $63a^2b$

2. Evaluate $5^2 \times 12^0$

- A) 0
- B) 10
- ☒ C) 25
- D) 120

3. Simplify $3e + 4c \times 5d$

- A) $3e + 9cd$
- ☒ B) $3e + 20cd$
- C) $23cde$
- D) $35cde$

4. Expand $-4(y - 6)$

- A) $4y - 10$
- B) $4y - 24$
- C) $-4y - 10$
- ☒ D) $-4y + 24$

5. The formula for the surface area of cylinder is

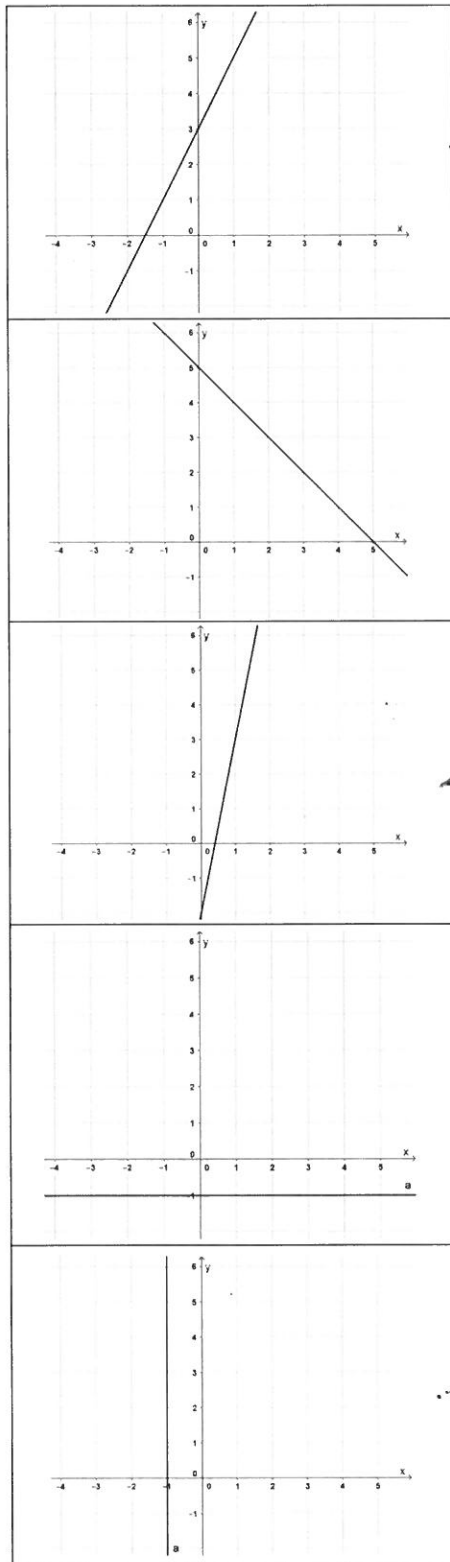
- A) $\pi r^2 h + 2\pi r$
- B) $2\pi r^2 + \pi r h$
- C) $2\pi r^2 h + 2\pi r$
- ☒ D) $2\pi r^2 + 2\pi r h$



r: radius
h: height

Matching Section: Draw a line to match the boxes on the left to the its corresponding box on the right

6. Match the graph with its linear equation



$$y = 5x - 2$$

$$y = 2x + 3$$

$$x = -1$$

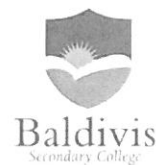
$$y = 5 - x$$

$$y = -1$$

Year 9 Mathematics Test

End of Term Test- Term 1

Resource Section (Calculators and notes allowed)



Time: 40 mins Name: _____ /40 marks

1. Use the index laws to simplify the following;

[8 marks]

a) $12^5 \div 12^3$ 12^2

b) $10^6 \times 10^0$ 10^6

c) $4h^{-2} \times 3h^5$ $12h^3$

d) $(g^3)^7$ g^{21}

e) $\frac{21a^5}{56a^2}$ $\frac{3a^3}{8}$

f) $\frac{w^8 \times w^{-9}}{w^3 p}$ $\frac{1}{w^4 p}$

2. Express the following with positive indices;

[2 marks]

a) 5^{-3} $\frac{1}{5^3}$ or $\frac{1}{125}$

b) $\frac{2}{e^{-5}}$ $2e^5$

3. Simplify the following expressions;

[2 marks]

a) $7 + 4x - 9x$ $7 - 5x$

b) $4k^2 \times 7k^3$ $28k^5$

4. Expand and simplify (where needed) the following expressions;

[4 marks]

a) $7(t-8)$ $7t - 56$

b) $-6m(4m+5)$
 $= -24m^2 - 30m$

c) $-p(2p+1) + 3p^2$
 $-2p^2 - p + 3p^2$
 $= p^2 - p$ //

5. Fully factorise the following expressions;

[4 marks]

a) $4x + 16$ $4(x+4)$

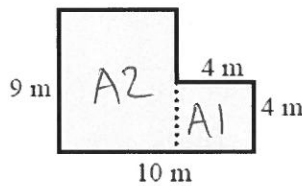
b) $15a - 3ab$
 $3a(5-b)$

c) $-24j^3 - 18j^2$
 $-6j^2(4j+3)$ ✓✓

6. Calculate the area of the following shapes;

[4 marks]

a)



$$A1 = 4 \times 4 = 16 \text{ m}^2$$

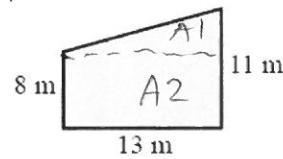
$$A2 = 9 \times 6 = 54 \text{ m}^2$$

$$A_T = A1 + A2$$

$$= 16 + 54$$

$$= 70 \text{ m}^2$$

b)



$$A1 = 3 \times 13 \times \frac{1}{2} = 19.5 \text{ m}^2$$

$$A2 = 8 \times 13 = 104 \text{ m}^2$$

$$A_T = A1 + A2$$

$$= 19.5 + 104$$

$$= 123.5 \text{ m}^2$$

7. Calculate the total surface area of the following shapes;

[6 marks]

a) A box with a width of 30cm, a length of 40cm and a height of 1.2m

$$SA = l \times w \times 2 + l \times h \times 2 + h \times w \times 2$$

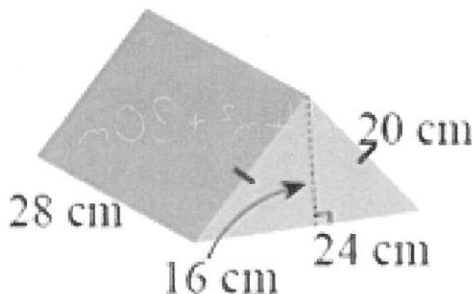
$$= 30 \times 40 \times 2 + 30 \times 120 \times 2 + 120 \times 30 \times 2$$

$$= 2400 + 9600 + 7200$$

$$= 19200 \text{ cm}^2$$

$$= 1.9 \text{ m}^2$$

b)



$$SA = 2 \times \frac{1}{2} \times 28 \times 24 + 2 \times 28 \times 16 + 24 \times 16$$

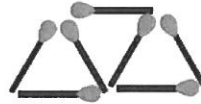
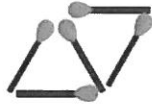
$$= 2 \times 24 \times 16 \times \frac{1}{2} + 2 \times 20 \times 28 + 24 \times 28$$

$$= 384 + 1120 + 672$$

$$= 2176 \text{ cm}^2$$

8. Consider the following pattern of matchstick arrangements.

[4 marks]



a) Is this pattern linear? Why/Why not?

Yes, it is linear because it goes up by the same amount

b) Fill in the table for the pattern if s is the shape number and m is the number of matches used. *each time*

s	1	2	3	4	5
m	3	5	7	9	11

c) Write a rule for the pattern above to connect the shape number to the number of matches used and use this rule to work out how many matches will be used for the 10th shape

$$m = 2s + 1$$

$$m = 2(10) + 1$$

$$m = 21$$

9. For the graph to the right;

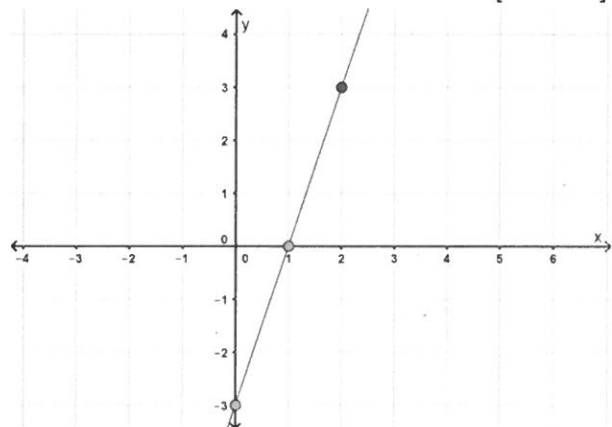
[2 marks]

a) State the gradient

$$\frac{3}{1} = 3$$

b) Use this gradient, along with the y-intercept, to create a linear equation for this graph in the form $y = mx + c$

$$y = 3x - 3$$

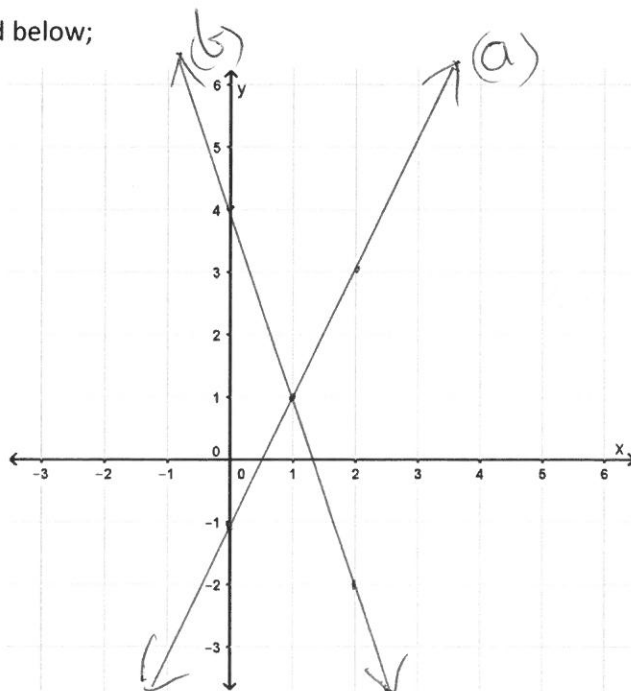


10. Sketch the following graphs on the grid below;

[4 marks]

a) $y = 2x - 1$

b) $y = -3x + 4$



Extension Question

1. Morgan is using a Pringles container (see picture) for a project but wants to cover it in coloured paper so it looks like a tower. Morgan thinks it'd be a good idea to turn it upside down so that way the lid isn't needed.

- a) If the tub is 7cm across and 23 cm high, what is the total area that needs to be covered?

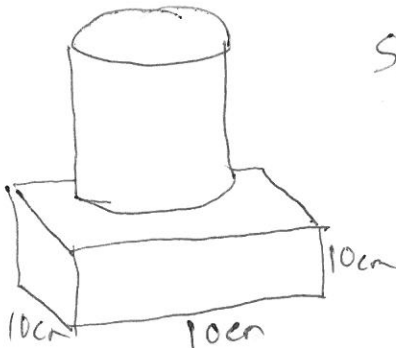
Hint: Don't forget, the lid isn't

[3 marks]
4

$$\begin{aligned}
 SA &= \pi r^2 + 2\pi r h \\
 &= \pi \times 3.5^2 + 2 \times \pi \times 3.5 \times 23 \\
 &= 38.48 + 505.8 \\
 &= 544.28 \text{ cm}^2
 \end{aligned}$$



- b) For the project, the tub is placed on a cube and Morgan thinks the cube should be the same colour. If the cube has a width of 10cm, what additional area needs to be covered? (Hint: the bottom of the cube doesn't need to be covered and only some of the top needs to be covered)



$$\begin{aligned}
 SA_{\text{cube}} &= L \times L \times 4 + (L \times L - \pi r^2) \quad [5 \text{ marks}] \\
 &= 10^2 \times 4 + (10^2 - \pi \times 3.5^2) \\
 &= 400 + (100 - 38.48) \\
 &= 461.52 \text{ cm}^2
 \end{aligned}$$

- c) Morgan works out the area of the sheets of coloured paper that will be used and thinks that all that needs to be done is to find the total surface area that needs to be covered and divide it by the area of the sheets to work out how many sheets will be needed. Explain why Morgan will actually need more paper than this.

[2 marks]
1

This is because there will be leftover paper